



HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

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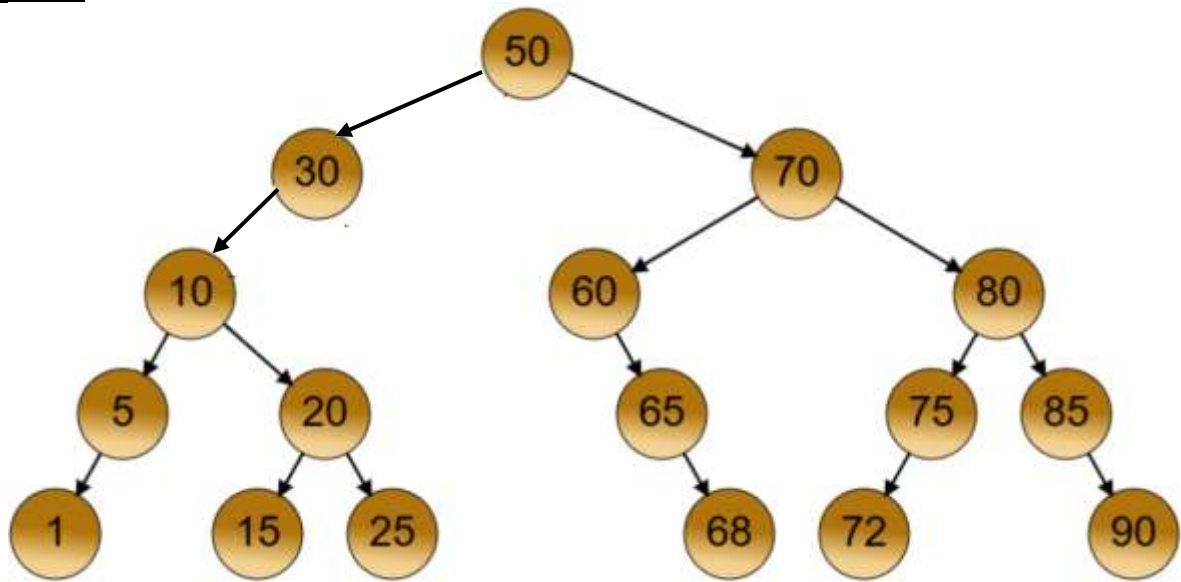
Quiz # 5 Solution

QUESTION: Binary Search Tree

1. For the given set of nodes, construct a binary search tree:

- Nodes = {50, 30, 70, 10, 60, 80, 5, 20, 65, 75, 85, 1, 15, 25, 68, 72, 90}

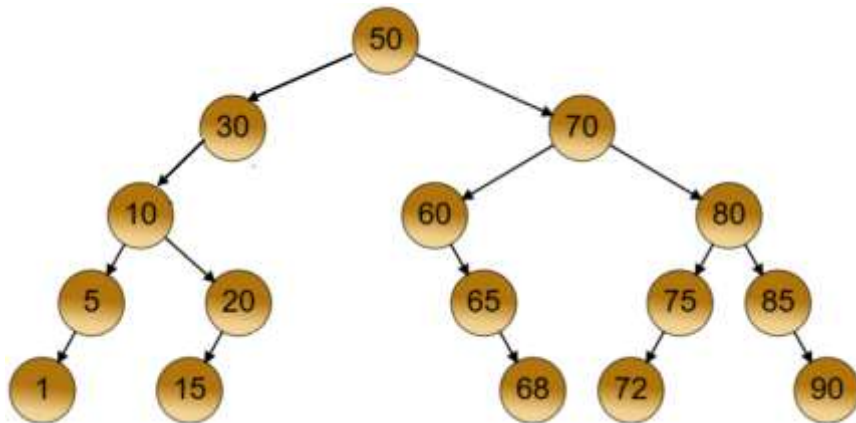
Resultant BST:



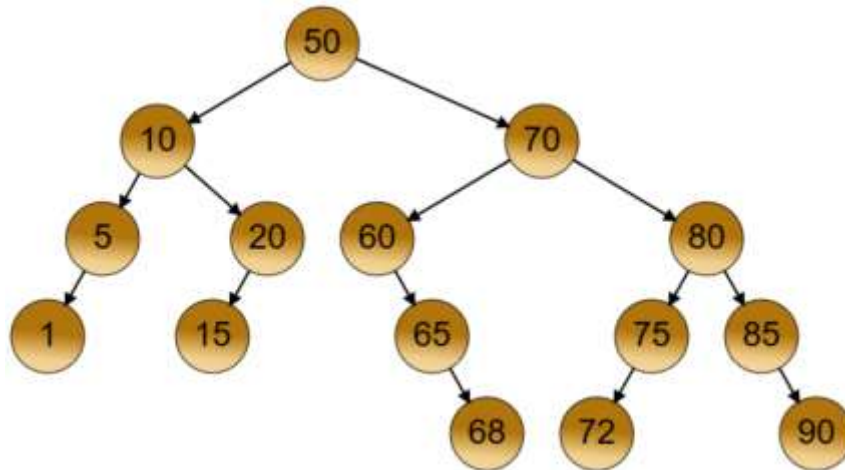
2. Search for Node 69 in the BST constructed in part (1), by specifying its complete path including its directions:
- Root: 50 → Right: 70 → Left: 60 → Right: 65 → Right: 68 → Right: Null**
Therefore, Node 69 not found
3. Search for the maximum in the BST constructed in part (1), by specifying its complete path including its directions:
- Root: 50 → Right: 70 → Right: 80 → Right: 85 → Right: 90 → Right: Null**
Therefore, Node 90 is the maximum in the given BST
4. Find the successor of Node 70 in the BST constructed in part (1), by specifying its complete path including its directions:
- Node: 70 → Right: 80 → Left: 75 → Left: 72 → Left: Null**
 - Therefore, inorder successor of Node 70 is Node 72**
5. Find the predecessor of Node 72 in the BST constructed in part (1), by specifying its complete path including its directions:
- Node: 72 → Parent: 75 → Parent: 80 → Parent: 70**
 - Node 70 has an immediately smaller value than that of Node 72, so that is its inorder predecessor**

6. Delete the following nodes one after the other from the BST constructed in part (1). Redraw the tree to get complete grade for deletion steps. Use inorder predecessor wherever needed for deletion.

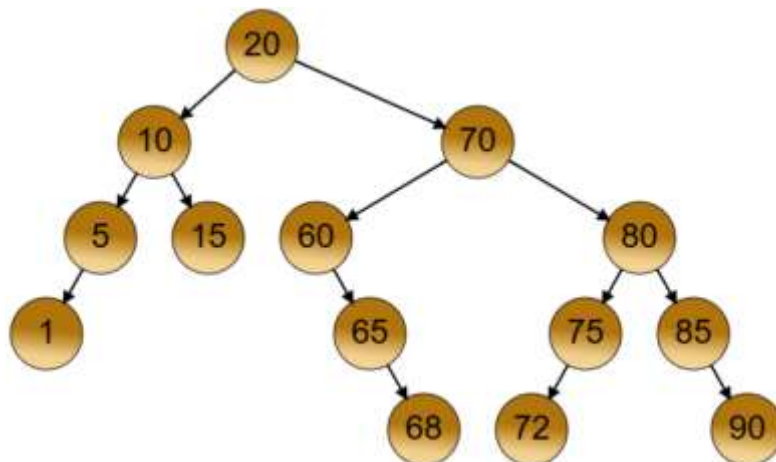
- Node 25: **Node 25 is a leaf node, so it can directly be removed from the tree, to get the resultant BST as follows:**



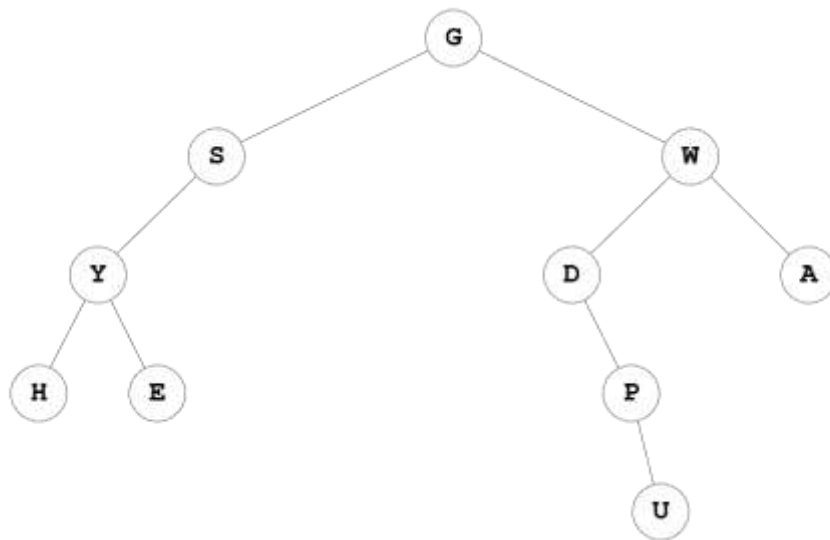
- Node 30: **Node 30 has one left child, so connect the left subtree to 30's parent, and get following BST:**



- Node 50: **Node 50 has two children. According to definition, we can replace it with its inorder successor or inorder predecessor, but as the question explicitly asks to replace with its predecessor, we will find its predecessor first**
 - Node: 50 → Left: 10 → Right: 20 → Right: Null**
 - Hence, 20 is the predecessor of 50, so replace 50 with it to get following resultant BST:**



7. For the tree given below, write down its Linked Structure Representation:



- { Root : G ,
Left : {
Root : S,
Left : {
Root : Y,
Left : H,
Right : E
},
Right : None
},
Right : {
Root : W,
Left : {
Root : D,
Left : None,
Right : {
Root : P,
Left : None,
Right : U
}
},
Right : A
}
}